Note-#Research

Date	March 17, 2026
Lecture	Research

ECON4200 (1)

📝 Summary

- Key Takeaways:
- Keywords:
- Important Subjects:

Persons Engaged and Job Vacancies in Establishments

- Period
  - 2000-2026
- Description
  - By industry
  - Establishment
    - A working unit - one kind of economic activity at a single physical location
  - Total
    - Engagement
    - Vancancy

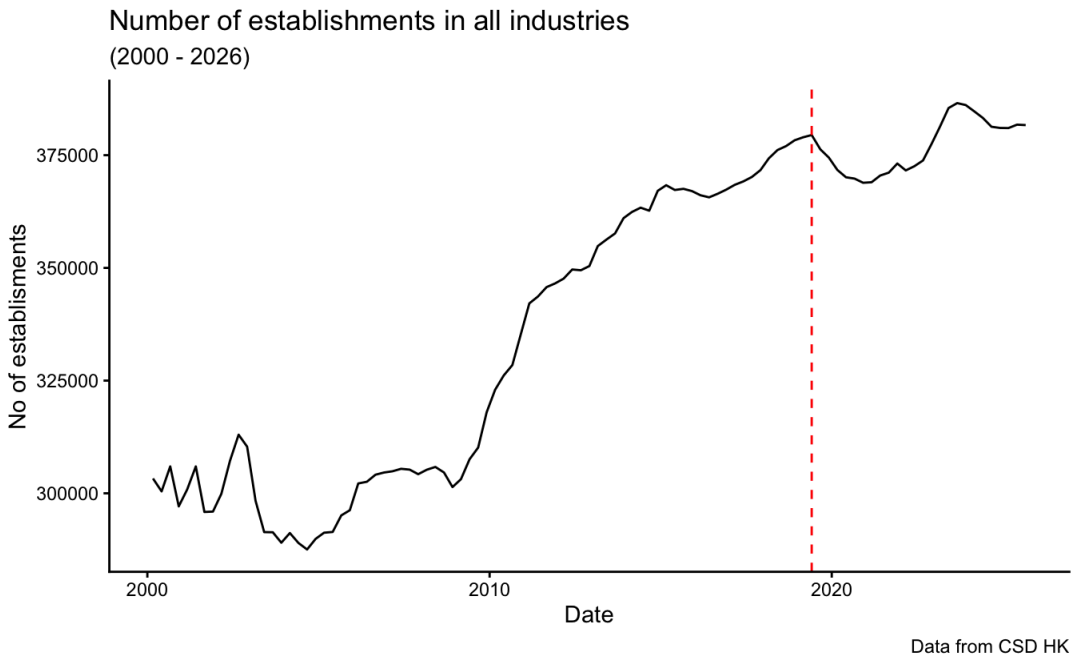
EDA

Insights

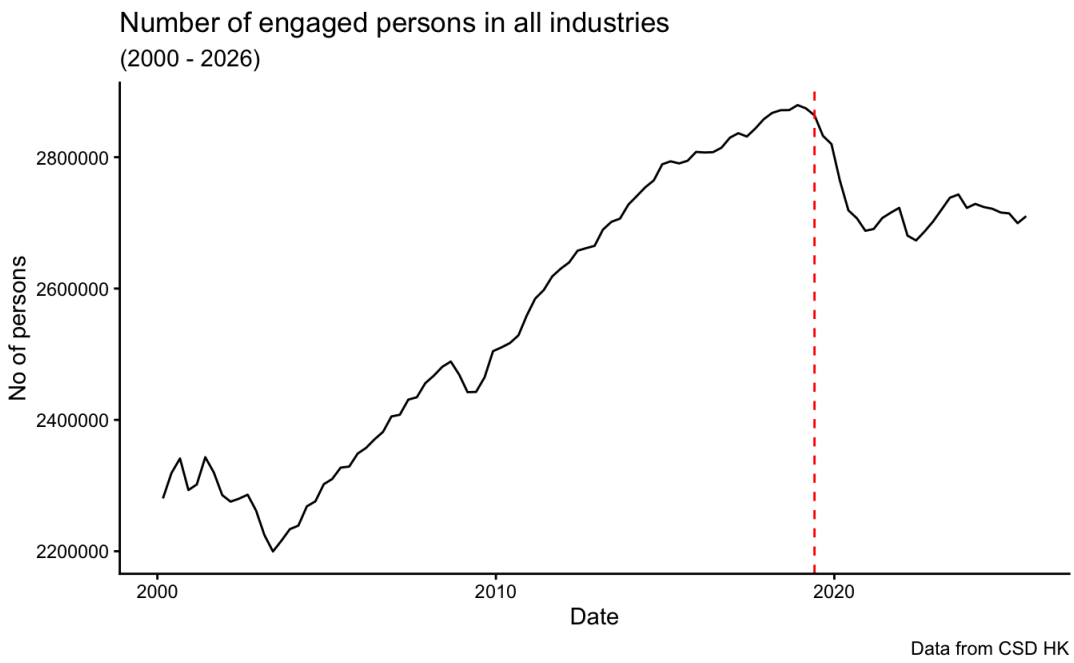
1. Engaged person decreased since 2019 and stop increasing since 2020
2. Establishments also decreased at 2019, but then keep increasing
3. Headcounts (mostly engaged + vacancies) per establishment decreased since 2009, flat til 2019, further decreased
4. Linear-log relationship: establishment / engaged person vs vacancies
5. Vacancies capped at 80k / 3%
6. Lead-lag relationship: engaged / vacancies dropped before establishments

▼ Totals

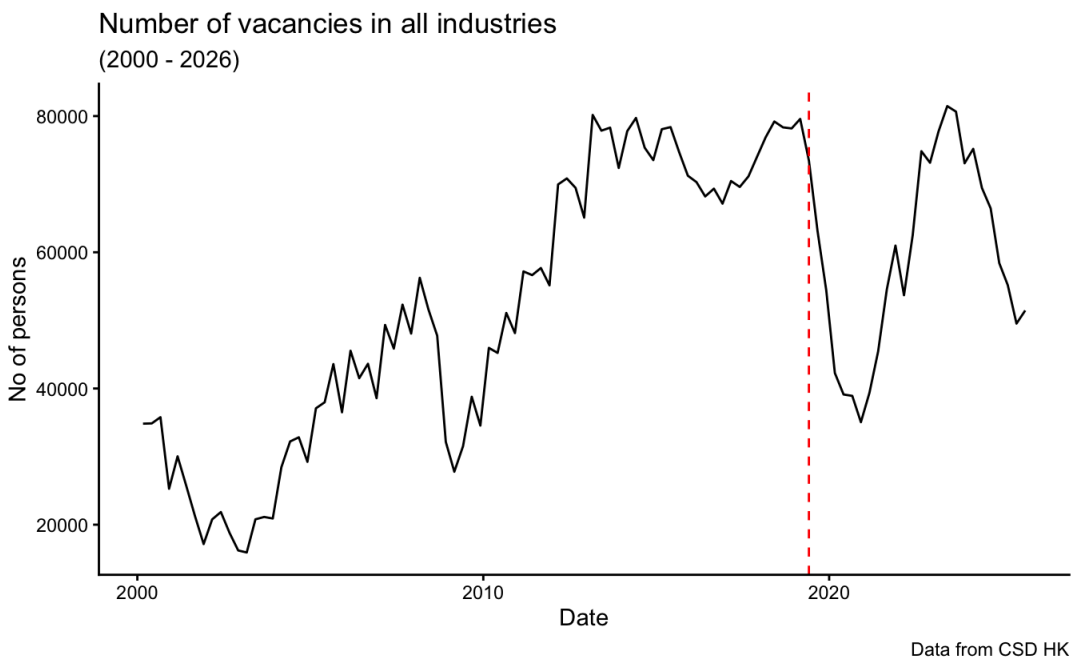
- Trend
  - Establishments trends up but plateau

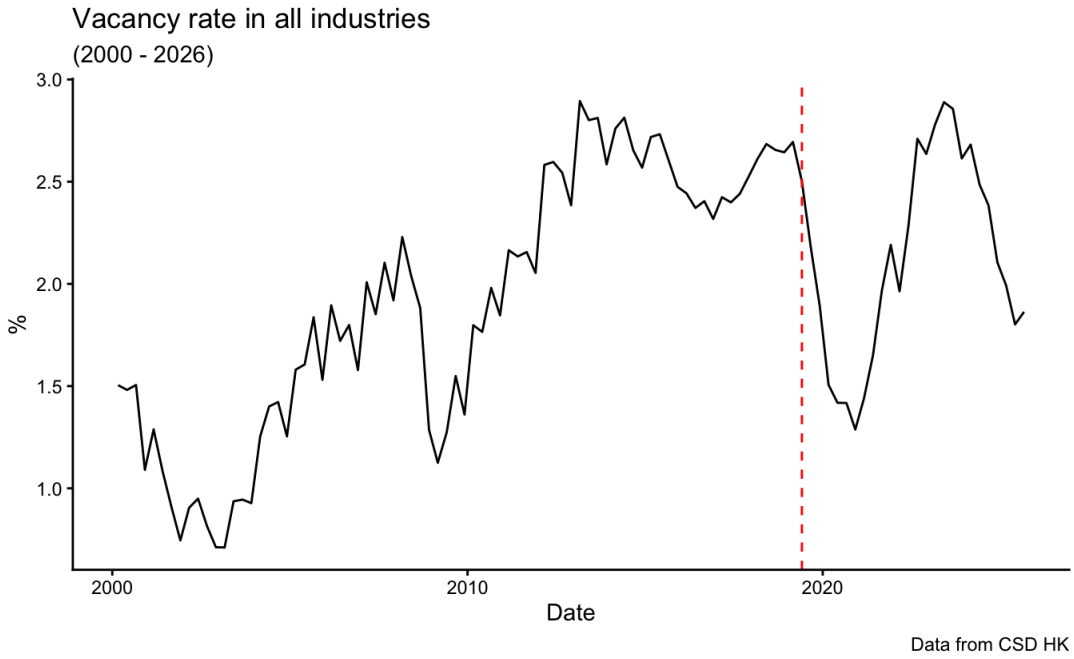


  - Engaged persons also plateau

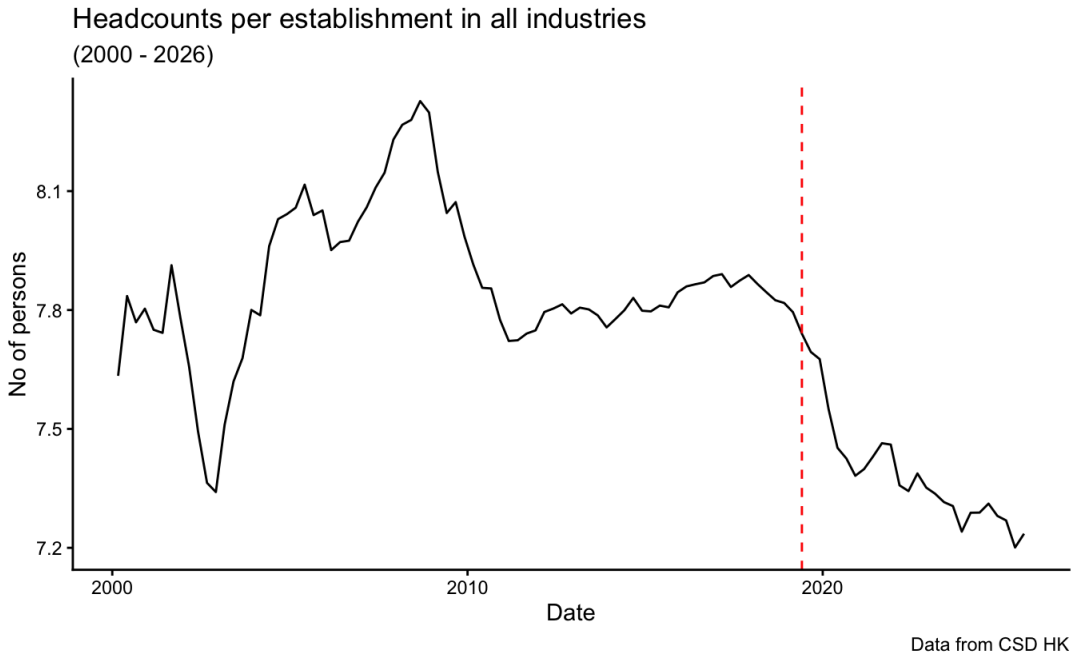


- Vacancies (rate) remains in a stabler range

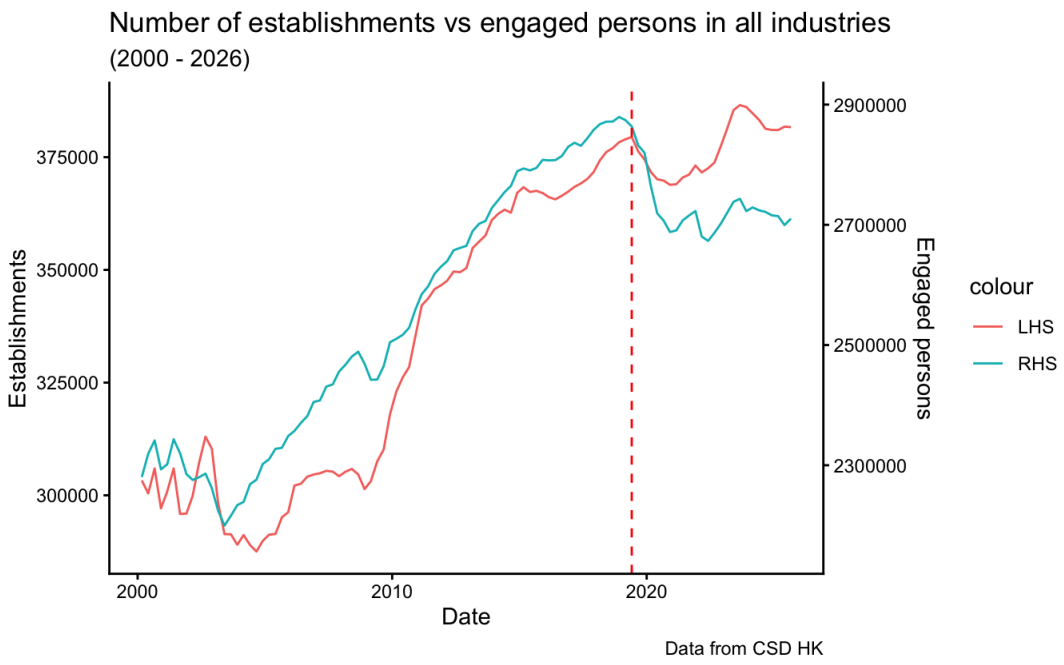




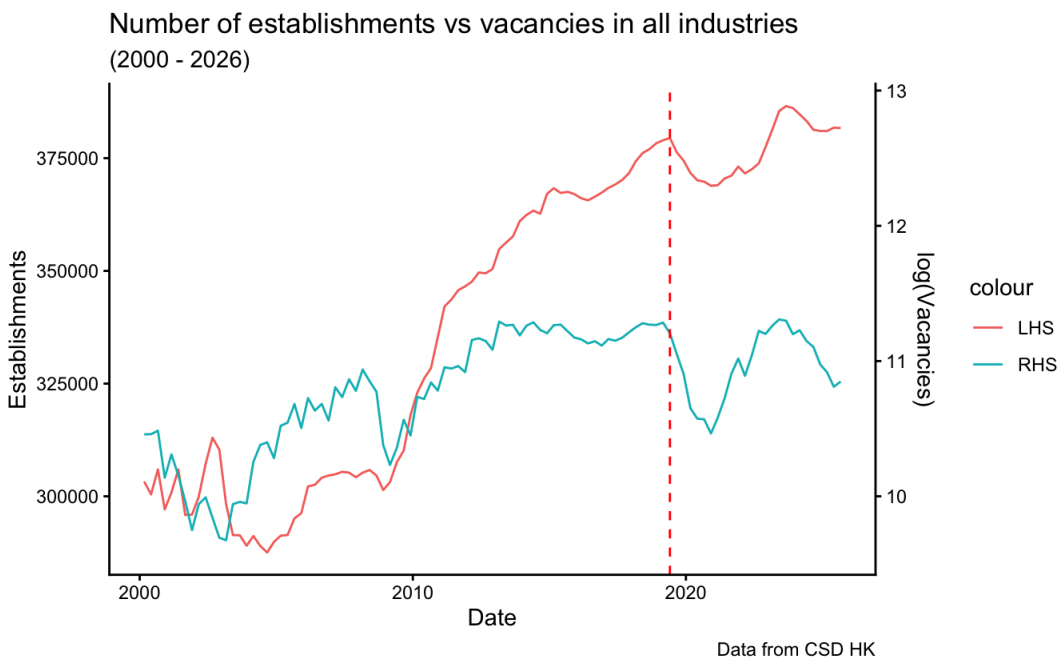
- <u>Headcounts / establishment declined after 2019</u>



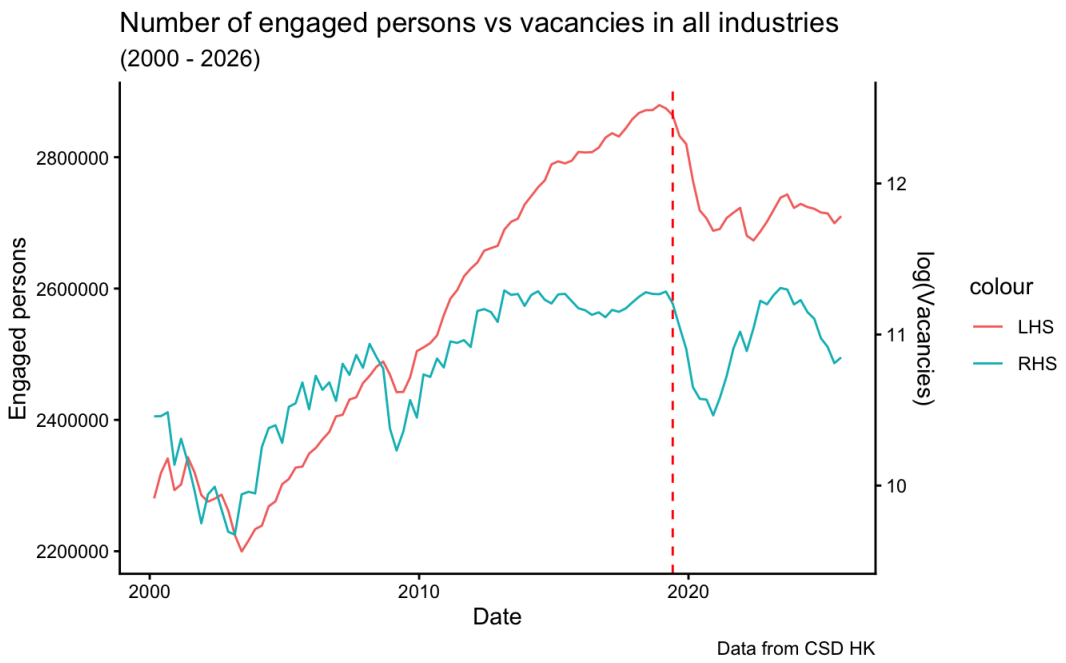
- Establishment vs engagement



- Establishment vs log(vacancies)

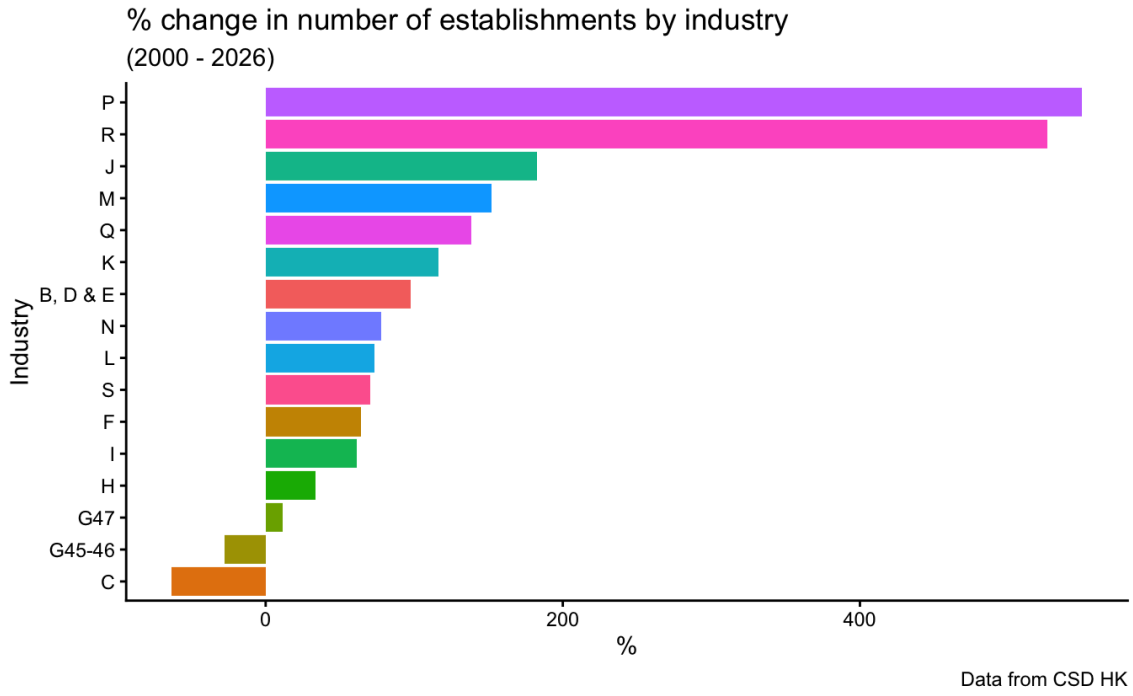


- Engagement vs log(vacancies)



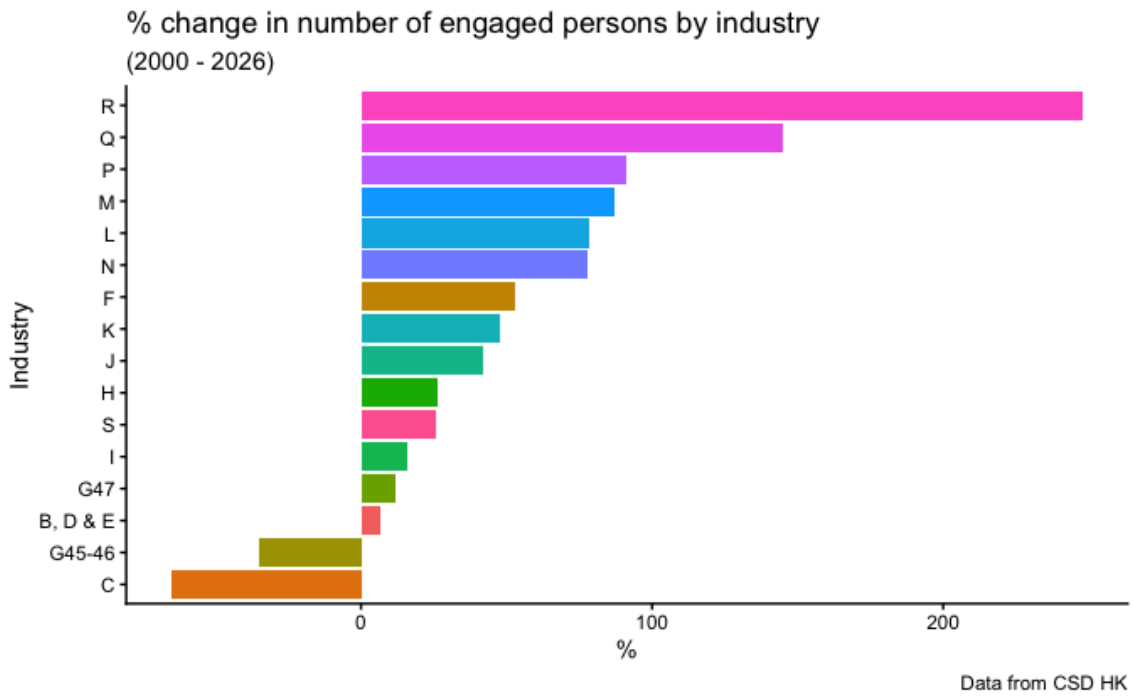
▼ By industry

- % changes
  - Establishment generally increased (edu, arts), except (trade, manufacturing)



	ind_des	change
	<fct>	<dbl>
1	P: Education	550.
2	R: Arts, entertainment and recreation	526.
3	J: Information and communications	182.
4	M: Professional, scientific and technical services	152.
5	Q: Human health and social work services	138.
6	K: Financing and insurance	116.
7	B, D & E: Mining and quarrying; and electricity and gas supply, and...	97.4
8	N: Administrative and support services	78.0
9	L: Real estate	73.2
10	S: Other services	70.5
11	F: Construction sites (manual workers only)	64.2
12	I: Accommodation and food services	61.1
13	H: Transportation, storage, postal and courier services	33.7
14	G47: Retail	11.4
15	G45-46: Import/export trade and wholesale	-27.4
16	C: Manufacturing	-63.2

- Engaged persons generally increased (art, social work, education), except (trade, manufacturing), patterns aligned



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   ind_des                                                                  change
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 1 R: Arts, entertainment and recreation                                    248.
 2 Q: Human health and social work services                                 145.
 3 P: Education                                                              90.9
 4 M: Professional, scientific and technical services                        86.8
 5 L: Real estate                                                            78.5
 6 N: Administrative and support services                                    77.7
 7 F: Construction sites (manual workers only)                               52.8
 8 K: Financing and insurance                                                47.8
 9 J: Information and communications                                         41.9
10 H: Transportation, storage, postal and courier services                   26.3
11 S: Other services                                                         25.5
12 I: Accommodation and food services                                        16.1
13 G47: Retail                                                               11.6
14 B, D & E: Mining and quarrying; and electricity and gas supply, and...     6.56
15 G45-46: Import/export trade and wholesale                               -35.2
16 C: Manufacturing                                                        -65.0
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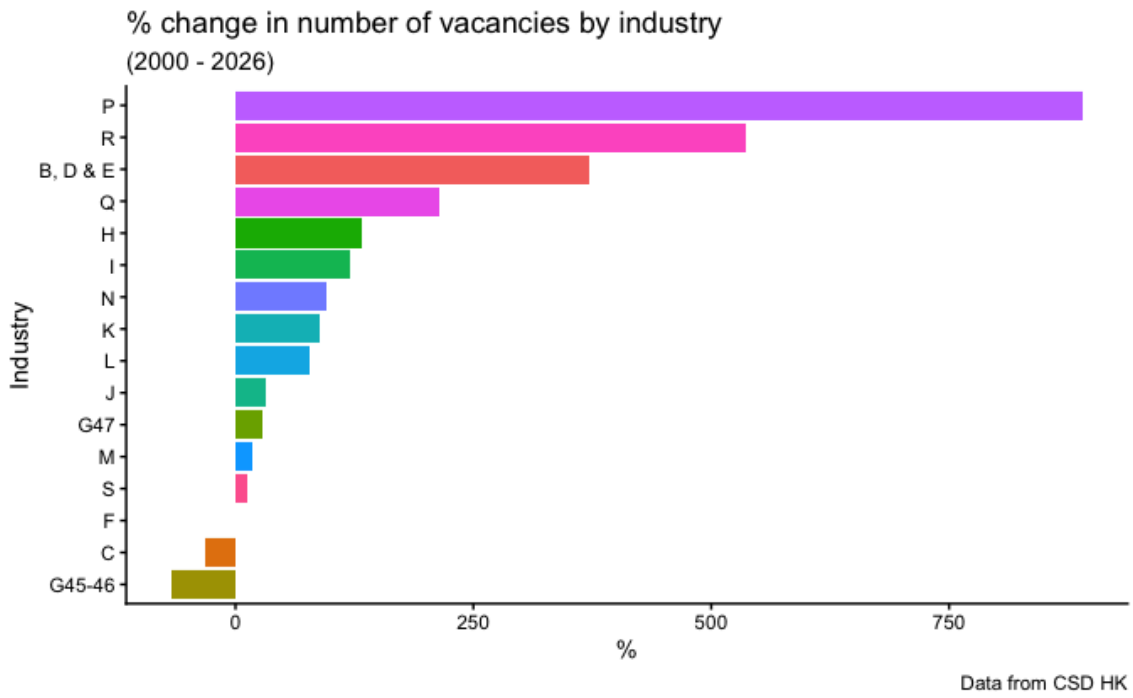
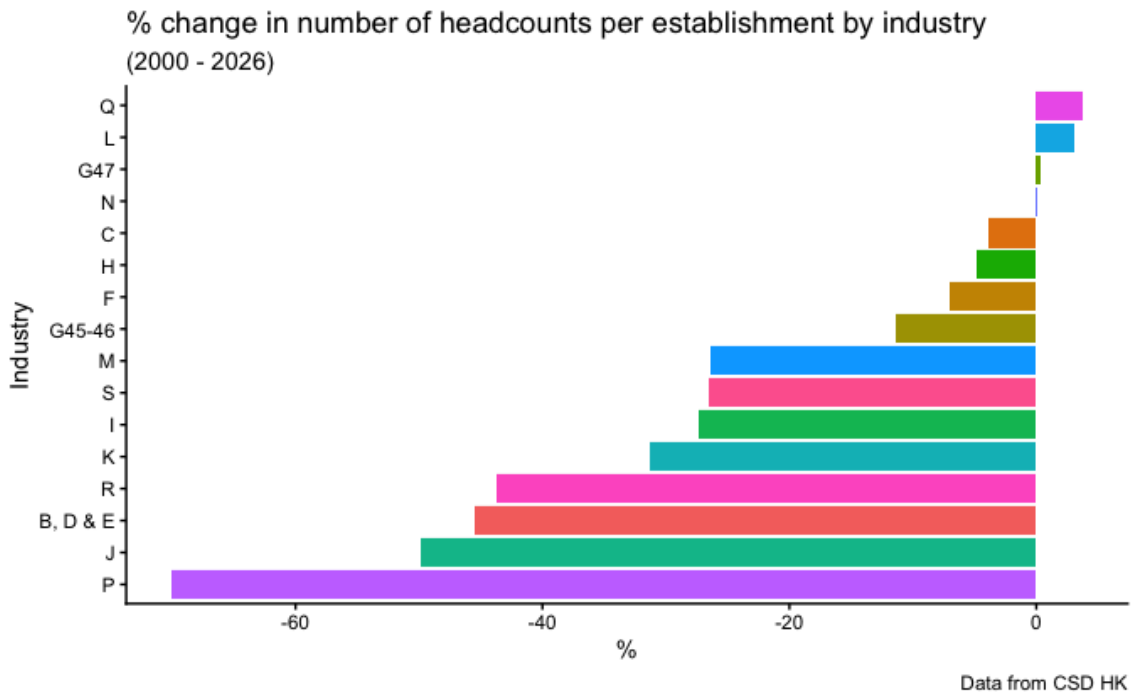
- Vacancy generally increased (edu, arts, mining, soc work), except (manufacturing, trade)



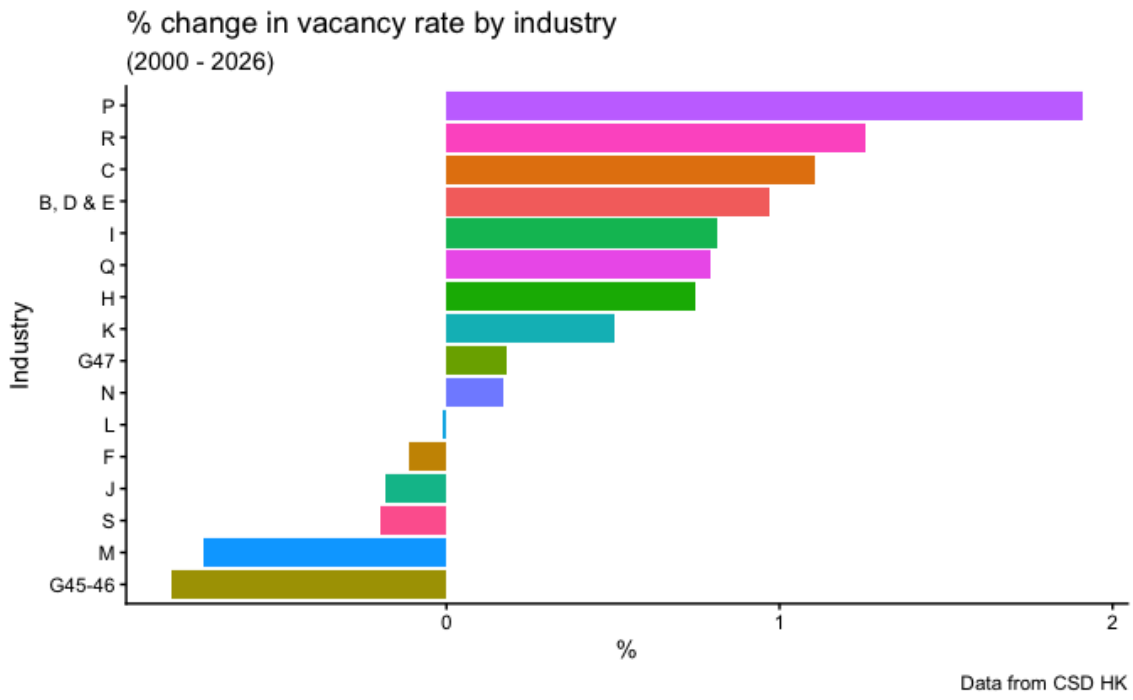
	ind_des	change
	<fct>	<dbl>
1	P: Education	890.
2	R: Arts, entertainment and recreation	535.
3	B, D & E: Mining and quarrying; and electricity and gas supply, and...	371.
4	Q: Human health and social work services	215.
5	H: Transportation, storage, postal and courier services	133.
6	I: Accommodation and food services	120.
7	N: Administrative and support services	95.3
8	K: Financing and insurance	88.2
9	L: Real estate	77.6
10	J: Information and communications	31.4
11	G47: Retail	28.7
12	M: Professional, scientific and technical services	17.4
13	S: Other services	11.7
14	F: Construction sites (manual workers only)	0
15	C: Manufacturing	-31.1
16	G45-46: Import/export trade and wholesale	-67.0

- Headcounts per establishment generally decreased (edu, info, mining, arts), except (soc work, real estate)



	ind_des	change
	<fct>	<dbl>
1	Q: Human health and social work services	3.71
2	L: Real estate	3.07
3	G47: Retail	0.410
4	N: Administrative and support services	0.0153
5	C: Manufacturing	-3.86
6	H: Transportation, storage, postal and courier services	-4.82
7	F: Construction sites (manual workers only)	-7.07
8	G45-46: Import/export trade and wholesale	-11.4
9	M: Professional, scientific and technical services	-26.5
10	S: Other services	-26.6
11	I: Accommodation and food services	-27.3
12	K: Financing and insurance	-31.2
13	R: Arts, entertainment and recreation	-43.8
14	B, D & E: Mining and quarrying; and electricity and gas supply, and...	-45.5
15	J: Information and communications	-49.9
16	P: Education	-70.0

- Vacancy rate generally increase, except (real estate, construction, info, professional, trade)



	ind_des	change
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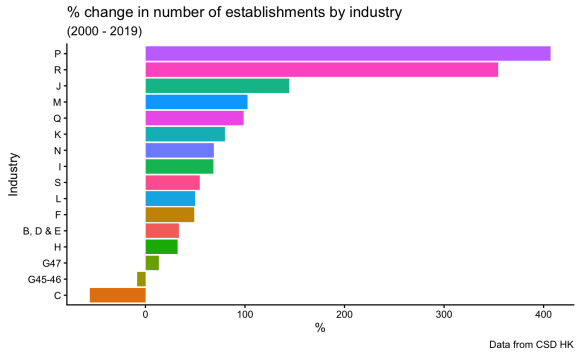
1	P: Education	1.91
2	R: Arts, entertainment and recreation	1.26
3	C: Manufacturing	1.11
4	B, D & E: Mining and quarrying; and electricity and gas supply, and...	0.971
5	I: Accommodation and food services	0.811
6	Q: Human health and social work services	0.793
7	H: Transportation, storage, postal and courier services	0.746
8	K: Financing and insurance	0.505
9	G47: Retail	0.182
10	N: Administrative and support services	0.169
11	L: Real estate	-0.0117
12	F: Construction sites (manual workers only)	-0.111
13	J: Information and communications	-0.181
14	S: Other services	-0.198
15	M: Professional, scientific and technical services	-0.730
16	G45-46: Import/export trade and wholesale	-0.825

Pre 2019

- Establishment

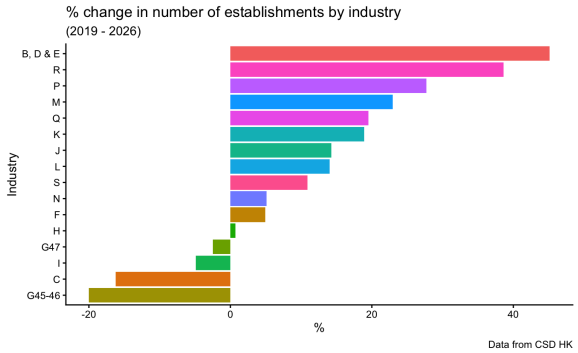


	ind_des	change
	<fct>	<dbl>
1	P: Education	407.
2	R: Arts, entertainment and recreation	354.
3	J: Information and communications	144.
4	M: Professional, scientific and technical services	103.
5	Q: Human health and social work services	98.7
6	K: Financing and insurance	79.9
7	N: Administrative and support services	68.9
8	I: Accommodation and food services	68.1
9	S: Other services	54.4
10	L: Real estate	49.8
11	F: Construction sites (manual workers only)	48.9
12	B, D & E: Mining and quarrying; and electricity and gas supply, and...	33.7
13	H: Transportation, storage, postal and courier services	32.3
14	G47: Retail	13.6
15	G45-46: Import/export trade and wholesale	-8.44
16	C: Manufacturing	-55.8

- Engaged persons

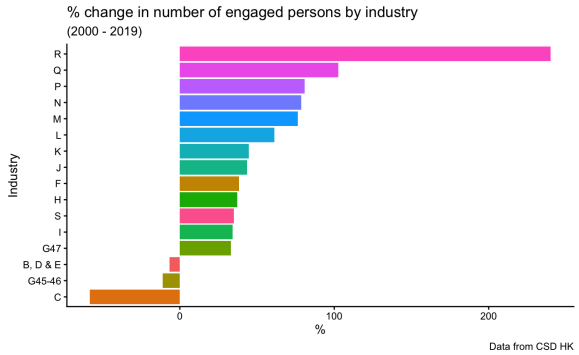
Post 2019

- Establishment

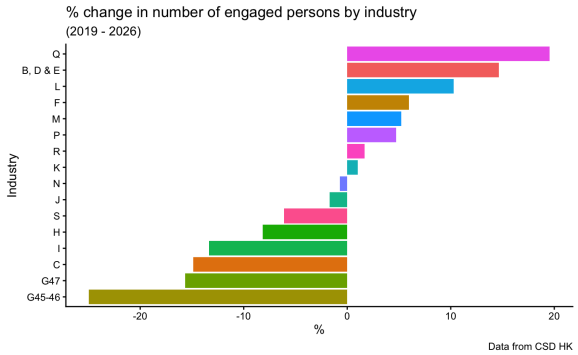


	ind_des	change
	<fct>	<dbl>
1	B, D & E: Mining and quarrying; and electricity and gas supply, and...	45.1
2	R: Arts, entertainment and recreation	38.6
3	P: Education	27.7
4	M: Professional, scientific and technical services	23.0
5	Q: Human health and social work services	19.5
6	K: Financing and insurance	18.9
7	J: Information and communications	14.3
8	L: Real estate	14.0
9	S: Other services	10.9
10	N: Administrative and support services	5.13
11	F: Construction sites (manual workers only)	4.92
12	H: Transportation, storage, postal and courier services	0.743
13	G47: Retail	-2.50
14	I: Accommodation and food services	-4.89
15	C: Manufacturing	-16.2
16	G45-46: Import/export trade and wholesale	-20.0

- Engaged persons



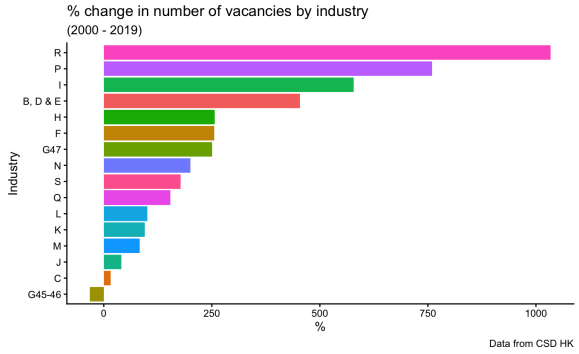
	ind_des	change
	<fct>	<dbl>
1	R: Arts, entertainment and recreation	240.
2	Q: Human health and social work services	103.
3	P: Education	81.0
4	N: Administrative and support services	78.7
5	M: Professional, scientific and technical services	76.3
6	L: Real estate	61.3
7	K: Financing and insurance	44.7
8	J: Information and communications	43.5
9	F: Construction sites (manual workers only)	38.2
10	H: Transportation, storage, postal and courier services	37.2
11	S: Other services	35.1
12	I: Accommodation and food services	34.2
13	G47: Retail	33.0
14	B, D & E: Mining and quarrying; and electricity and gas supply, and...	-6.60
15	G45-46: Import/export trade and wholesale	-11.1
16	C: Manufacturing	-58.2



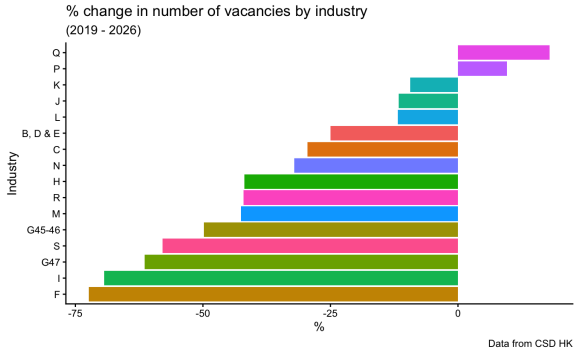
	ind_des	change
	<fct>	<dbl>
1	Q: Human health and social work services	19.6
2	B, D & E: Mining and quarrying; and electricity and gas supply, and...	14.7
3	L: Real estate	10.3
4	F: Construction sites (manual workers only)	5.95
5	M: Professional, scientific and technical services	5.23
6	P: Education	4.72
7	R: Arts, entertainment and recreation	1.68
8	K: Financing and insurance	1.01
9	N: Administrative and support services	-0.697
10	J: Information and communications	-1.72
11	S: Other services	-6.12
12	H: Transportation, storage, postal and courier services	-8.16
13	I: Accommodation and food services	-13.4
14	C: Manufacturing	-14.9
15	G47: Retail	-15.6
16	G45-46: Import/export trade and wholesale	-25.0

- Vacancies

- Vacancies



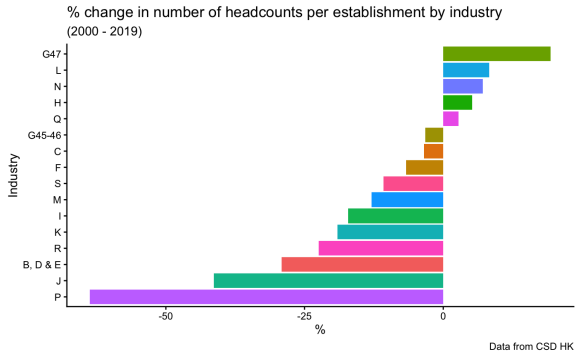
	ind_des	change
	<fct>	<dbl>
1	R: Arts, entertainment and recreation	1034.
2	P: Education	760.
3	I: Accommodation and food services	578.
4	B, D & E: Mining and quarrying; and electricity and gas supply, and...	454.
5	H: Transportation, storage, postal and courier services	257.
6	F: Construction sites (manual workers only)	255.
7	G47: Retail	250.
8	N: Administrative and support services	201.
9	S: Other services	178.
10	Q: Human health and social work services	154.
11	L: Real estate	101.
12	K: Financing and insurance	95.2
13	M: Professional, scientific and technical services	83.4
14	J: Information and communications	40.3
15	C: Manufacturing	15.5
16	G45-46: Import/export trade and wholesale	-32.1

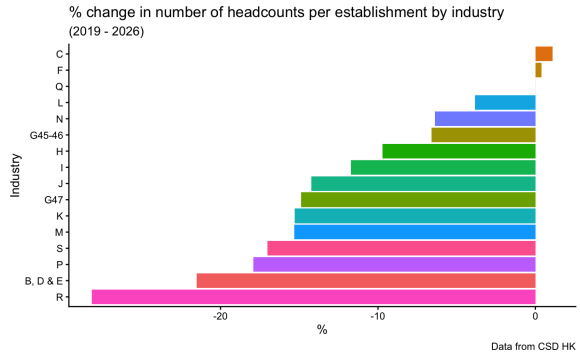


	ind_des	change
	<fct>	<dbl>
1	Q: Human health and social work services	18.0
2	P: Education	9.57
3	K: Financing and insurance	-9.39
4	J: Information and communications	-11.6
5	L: Real estate	-11.8
6	B, D & E: Mining and quarrying; and electricity and gas supply, and...	-25
7	C: Manufacturing	-29.5
8	N: Administrative and support services	-32.1
9	H: Transportation, storage, postal and courier services	-41.9
10	R: Arts, entertainment and recreation	-42.0
11	M: Professional, scientific and technical services	-42.6
12	G45-46: Import/export trade and wholesale	-49.8
13	S: Other services	-57.9
14	G47: Retail	-61.4
15	I: Accommodation and food services	-69.4
16	F: Construction sites (manual workers only)	-72.4

- Headcounts

- Headcounts





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1	G47: Retail	19.4
2	L: Real estate	8.30
3	N: Administrative and support services	7.11
4	H: Transportation, storage, postal and courier services	5.18
5	Q: Human health and social work services	2.75
6	G45-46: Import/export trade and wholesale	-3.26
7	C: Manufacturing	-3.48
8	F: Construction sites (manual workers only)	-6.69
9	S: Other services	-10.8
10	M: Professional, scientific and technical services	-12.9
11	I: Accommodation and food services	-17.2
12	K: Financing and insurance	-19.0
13	R: Arts, entertainment and recreation	-22.4
14	B, D & E: Mining and quarrying; and electricity and gas supply, and...	-29.1
15	J: Information and communications	-41.3
16	P: Education	-63.7

	ind_des	chang
	<fct>	<dbl
1	C: Manufacturing	1.10
2	F: Construction sites (manual workers only)	0.379
3	Q: Human health and social work services	0.0052
4	L: Real estate	-3.83
5	N: Administrative and support services	-6.37
6	G45-46: Import/export trade and wholesale	-6.60
7	H: Transportation, storage, postal and courier services	-9.70
8	I: Accommodation and food services	-11.7
9	J: Information and communications	-14.2
10	G47: Retail	-14.9
11	K: Financing and insurance	-15.3
12	M: Professional, scientific and technical services	-15.3
13	S: Other services	-17.0
14	P: Education	-17.9
15	B, D & E: Mining and quarrying; and electricity and gas supply, and...	-21.5
16	R: Arts, entertainment and recreation	-28.2

✅To-Dos

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